A Socioeconomic Informative Dashboard for Migrations and Relocations

In partial fulfillment of the requirements for the course
Visualizations and Data Analysis
1st Semester, S.Y. 2025-2026

Submitted by:

Pangilinan Jr. Bienvenido B.
Pasion, Al Twain D.
Torres, Carl Andrei D.

Submitted to:

Engr. Jimlord Quejado

Technological Institute of the Philippines - Quezon City

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I. Problem Background and Validation

- The problem is present around the world, understanding it is critical to avoid misconception that a low-cost of living translates to an easy life. This validates the issue through the macroeconomic dataset and shifts our understanding about rent to objective reality, demonstrating how housing costs can dictate the overall quality of life of the nation's

citizens.

a. The Problem

- The project addresses the global housing affordability crisis, specifically the property price to income ratio. This project utilizes non-synthetic and international data to present that this crisis is “global”, a worldwide phenomenon where the economic prices of homeownership are increasing, severely degrading their overall quality of life.

- This problem affects young individuals and first-time buyers because they are just beginning to earn income to buy a new home. It also impacts low to middle-income families because, despite having jobs, their earnings are insufficient to save for their families' health, safety, and standard of living. Furthermore, the urban workforce is often forced into longer commutes and lower-quality working environments. On a broader scale, government officials and real estate developers are directly affected because severe housing unaffordability depresses national quality-of-life scores, threatens national economic growth, and signals real-estate market instability that forces real estate developers to alter how they operate.

WHO MIGHT BE AFFECTED BY THIS PROBLEM:

- **Young Individuals and First-Time Buyers** -> they are just beginning to earn income to buy a new home.
- **Low- to Middle-Income Families** -> Despite having jobs, their earnings are insufficient to save for their families' health, safety, and standard of living.
- **Urban Workforce** -> Often forced into longer commutes and lower-quality working environments.
- **Government Officials** -> Housing Affordability Crisis affects the severe housing unaffordability depresses national quality-of-life scores and the national economic growth.
- **Real Estate Developers** -> Housing Affordability Crisis signals real-estate market instability which forces real real estate developers to alter how they operate.

<Insert an image, diagram, or storyboard sketch depicting the problem scenario>

b. Problem Validation

- The problem is verified to exist, according to Yanotti et al. (2023), across many nations, the gap between house prices and household incomes has widened internationally. This demonstrates that housing affordability is a systemic, structural economic issue rather than a personal financial failing.

reference: https://doi.org/10.2139/ssrn.4402329

II. The Project

- This project utilizes statistical data visualization to map out the correlation between a nation’s housing affordability and its citizens’ overall quality of life. By using the effective graphs, the goal of this project is to visually rebrand the belief that moving to a country with a “low cost of living” can help with financial struggles. Thus proving that local purchasing power and housing raptors are one of the main factors of livability.

a. Dataset

ORIGIN OF THE DATASET:

The Dataset comes from Kaggle.com and the Dataset was from **Numbeo** which the link is:

https://www.kaggle.com/datasets/ahmedmohamed2003/quality-of-life-for-each-country

CONTRIBUTORS:

Ahmed Mohamed

TYPE OF DATA:

This dataset is composed of ***numerical scores*** and ***descriptive categories***.

Numerical Scores:
- Purchasing Power Value
- Safety Value
- Health Care Value
- Climate Value
- Cost of Living Value

Descriptive Categories:
- Purchasing Power Category
- Safety Category
- Health Care Category
- Climate Category

TOTAL NUMBER OF ENTRIES:
Total Rows: 236 Countries (each row represents each country)
Total Columns: 19 Attributes (1 entity identifier column, 9 numerical score columns, and 9 qualitative category columns)

RELEVANT ATTRIBUTES:

CORE ENTITY:

- **Country** -> name of a country

KEY ATTRIBUTES TO HOUSING AFFORDABILITY ANALYSIS:

- **Property Price to Income Value *(Numerical)*** -> Lower scores indicate better affordability; higher values mean housing is severely expensive relative to average earnings.
- **Property Price to Income Category *(Categorical)*** -> Qualitative rating (e.g., 'Low', 'Moderate', 'High', 'Very High').
- **Quality of Life Value *(Numerical)*** -> Combined index measuring overall living conditions.
- **Quality of Life Category *(Categorical)*** -> Qualitative rating of overall living standards.
- **Cost of Living Value / Cost of Living Category** -> Overall cost of goods and services relative to base indices.
- **Purchasing Power Value / Purchasing Power Category** -> Relative financial capacity of citizens.

SUPPORTING QUALITY-OF-LIFE ATTRIBUTES:

- **Safety Value & Safety Category** -> National safety and crime rate perception metrics.
- **Health Care Value & Health Care Category** -> Assessment of healthcare quality, responsiveness, and infrastructure.
- **Traffic Commute Time Value & Traffic Commute Time Category** -> Average time spent commuting (reflects urban sprawl and congestion).
- **Pollution Value & Pollution Category** -> Environmental cleanliness and pollution exposure index.
- **Climate Value & Climate Category** -> Climate favorability score.

<Mention cleaning, normalization, or transformation processes>

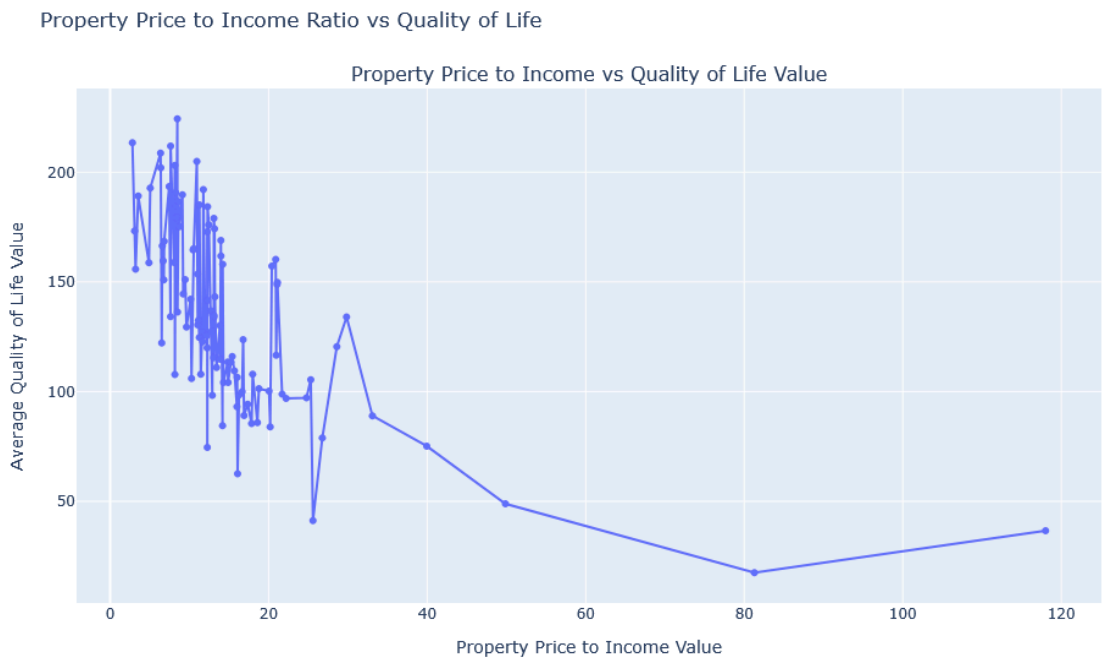
b. Data Dashboard
<Explain what the dashboard is for e.g., visualization, decision support, etc>
<List main dashboard components and functionalities>

c. Project Description
<Briefly describe the overall project or system>
<List the key objectives or deliverables>

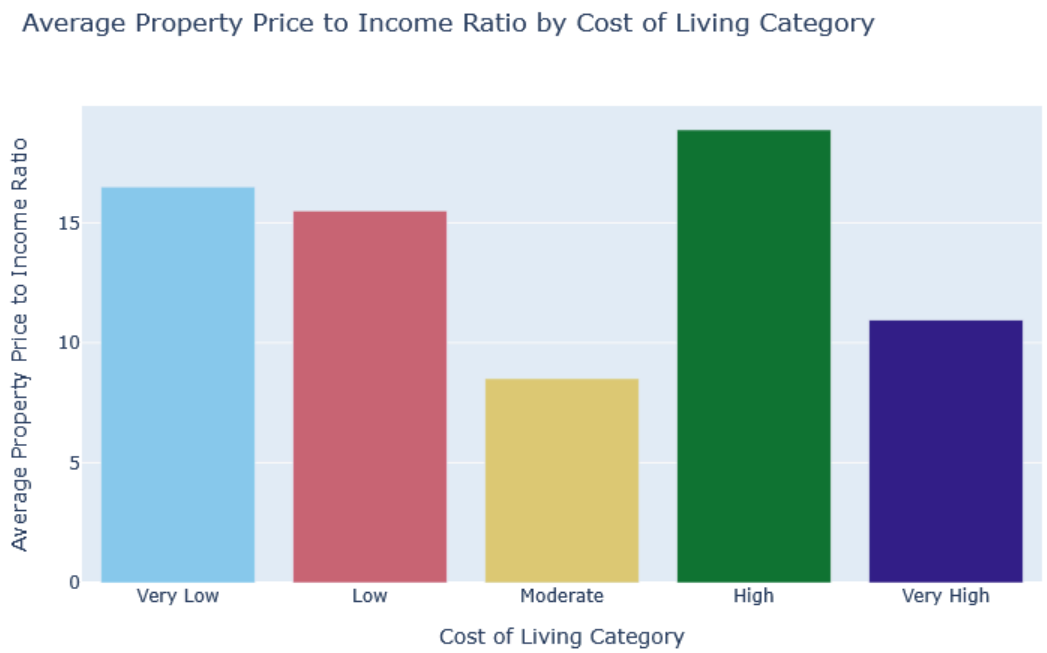
III. Output

<Insert screenshots, charts, tables, or figures showing the final product.>
<Summarize the most important outcomes of your project or analysis>
<Add storyboard panels that show before/after scenarios or end-user experience>

Property Price to Income Ratio VS Quality of Life Value



Average Property to Price Income Ratio by Cost of Living Category



V. Development Team

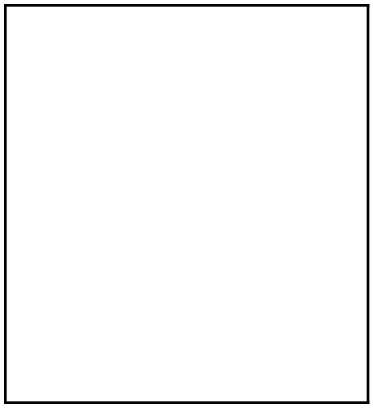
- Team 6 is composed of three students from the Technological Institute of the Philippines - Quezon City who developed this project for the course Visualizations and Data Analysis. The team studied the relationship between housing affordability and quality of life across countries using the Numbeo dataset from Kaggle. All three members took part in data collection, while the remaining tasks were divided according to each member's role, Pasion led the system design, and Pangilinan and Torres handled the slides, documentation, and presentation.

a. Team Description

Pangilinan	- Data collection, PPT, Documentation, Presenter
Pasion	- System design, Data collection
Torres	- Data collection, PPT, Documentation, Presenter

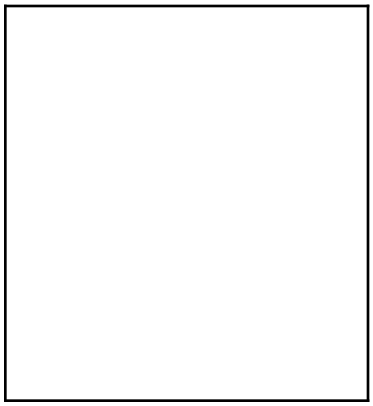
- The team works as a small group in which everyone contributes to the data and each member owns a specific part of the output. Pasion focuses on how the system and dashboard are designed, while Pangilinan and Torres focus on communicating the results through the slides, the written documentation, and the final presentation. The team's goal is to present clear, data-driven visualizations that show how housing affordability shapes a country's quality of life, and to challenge the belief that a low cost of living automatically means an easier life.

b. Profile



Pangilinan Jr. Bienvenido B.

- I am Bienvenido B. Pangilinan Jr., a student at the Technological Institute of the Philippines - Quezon City (TIP-QC) and a member of Team 6 in Visualizations and Data Analysis. In this project I took part in data collection and was responsible for the PPT, the documentation, and presenting the team's work. Skills and expertise: data collection, document writing, slide preparation, and presenting.



Pasion, Al Twain D.

- I am Al Twain D. Pasion, a college student at the Technological Institute of the Philippines - Quezon City (TIP-QC). I'm currently taking the Bachelor of Science in Computer Engineering program. I'm adequate in basic C++ and Python programming languages. I value the process of learning and believe that time is only secondary when it comes to what you're passionate about. In this project, I am the lead coder, data collector, presentation maker, and debugger.



Torres, Carl Andrei D.

- I am Carl Andrei D. Torres, a 2nd Year Student at Technological Institute of the Philippines - Quezon City (TIP-QC) where I serve as the Documenter, Data Collector, Presenter, and PPT Creator of our Storyboard in Visualizations and Data Analysis. I am interested in making documents and presentations (PPT) which I'm good at. Also, I am one of the Data Collectors for our Storyboard which can be helpful for understanding the solution to our problem carefully. Also, I am one of the presenters that will present our created PPT and Storyboard which can be crucial for explaining the solution that we've created for the problem. Moreover, I've managed to create documents and presentations in recent projects from different courses. Lastly, my goal for this project is to create documents and presentation that are understandable for the executive and manages to present for us to connect our solution to the problem successfully based from our storyboard output.